

Serotonin

1. From where in the body do we obtain Tryptophan? How does this relate to ATD?
2. Draw the enzymatic synthesis of Serotonin.
Note: names of precursors, enzymes, co-factors, etc. are sufficient. It is NOT necessary to draw the chemical structures.
3. Almost all 5-HT neurons in the CNS are found along the midline of the midbrain. Which 2 areas supply most of the CNS with 5-HT? What area supplies the spinal cord and hindbrain (cerebellum) with 5-HT?
4. There are two main classes of 5-HT autoreceptors. Compare and contrast these two classes and their roles in neurotransmission.

5-HT _{1B/1D} terminal autoreceptors	5-HT _{1A} somatodendritic autoreceptors

5. There are at least 15 different types of 5-HT receptors, named 5-HT₁ – 5-HT₇. Compare and contrast 5-HT_{1A} and 5-HT_{2A} receptors and their roles in neurotransmission.

5-HT _{1A}	5-HT _{2A}

6. There are two key forms of 5-HT inactivation. Compare and contrast both, including major metabolites, proteins involved, and pharmaceutical impacts.

Reuptake	Enzymatic Degradation

7. Explain the experimental evidence describing the role of MDMA in the 5-HT system relating to serotonergic dendritic spine density and PTSD treatment.

8. 5-HT neurotransmission controls _____ , with roles relating to aggression, anxiety, and feeding. The 5-HT Deficiency Hypothesis suggests that impaired 5-HTergic signaling increases behavioral aggression. What are the two main lines of evidence for this hypothesis? Do they support the hypothesis?

i.

ii.

9. What 3 5-HT receptors are shown to be implicated in anxiety?

i.

ii.

iii.

What about feeding?

i.

ii.